

DATA STRUCTURES PROJECT

Food menu

Restaurant Management system v1.0

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WHO WE ARE:

Name: Miqdad Osama

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Project: Food menu

The challenge

Modern restaurants require fast, reliable, and persistent ways to manage their menus. Traditional methods of data entry can be slow and prone to errors.

Our solution is a C++ based Management system that offers $O(1)$ search speeds and persistent file storage, ensuring data safety and system performance.



System Core Features

1. Implemented using linked list:

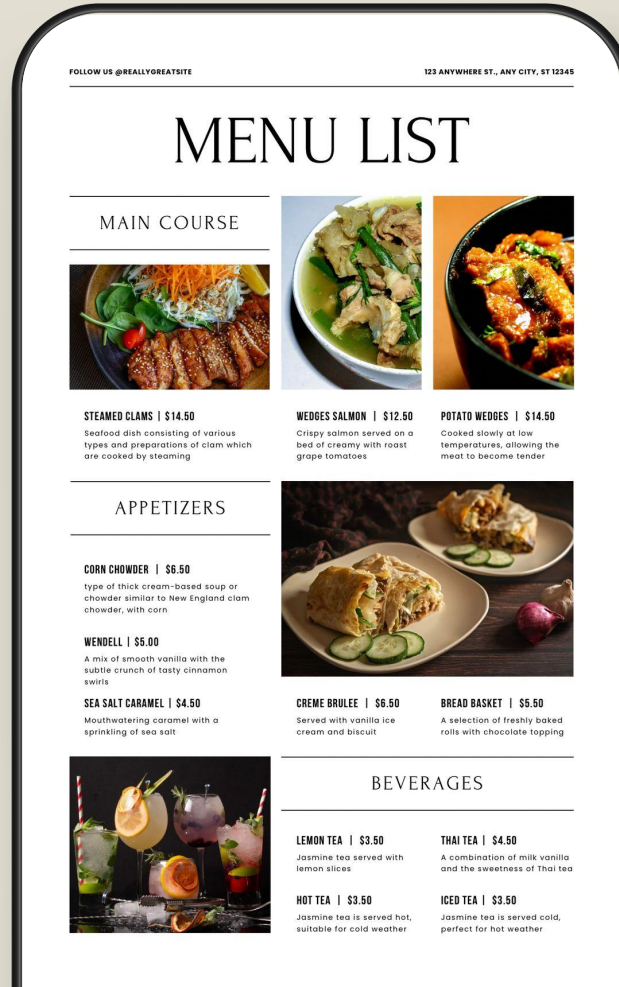
- To make the program more flexible when adding, editing, and deleting the meals.
- The reason for choosing the linked list is that the linked list is the best choice for performing adding, because the number may increase or decrease.

2. Implemented using stack:

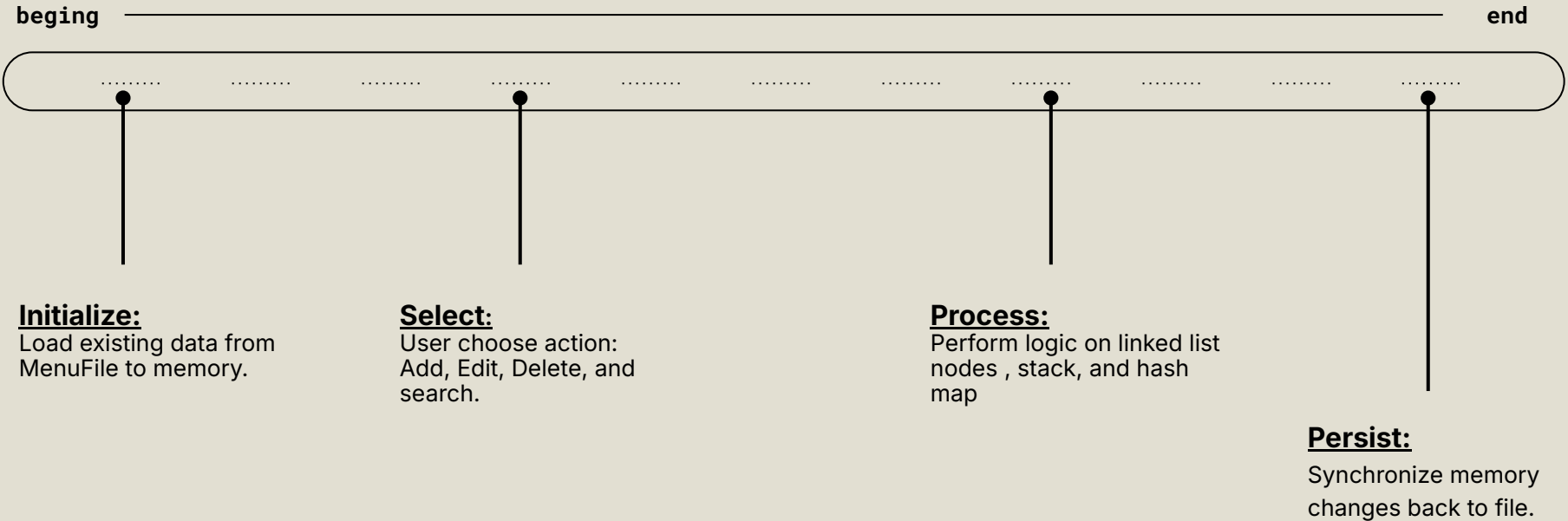
- If a user deletes something and wants to undo the deletion.
- The reason is that the stack is the best data structure for saving data temporarily.

3. Hash table:

- Because the search operation is very slow in linked lists, we use a hash table for faster access and making the user experience better.



Logic Flowchart

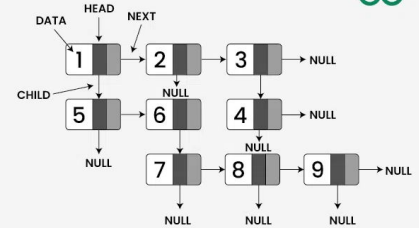


Linked list

The system utilizes a Custom Singly Linked List to store meal nodes dynamically. This allows for unlimited menu growth without memory reallocation penalties.

Each node contains a mealData pointer and a link to the next element, facilitating linear traversal and display.

Linked List of Linked List

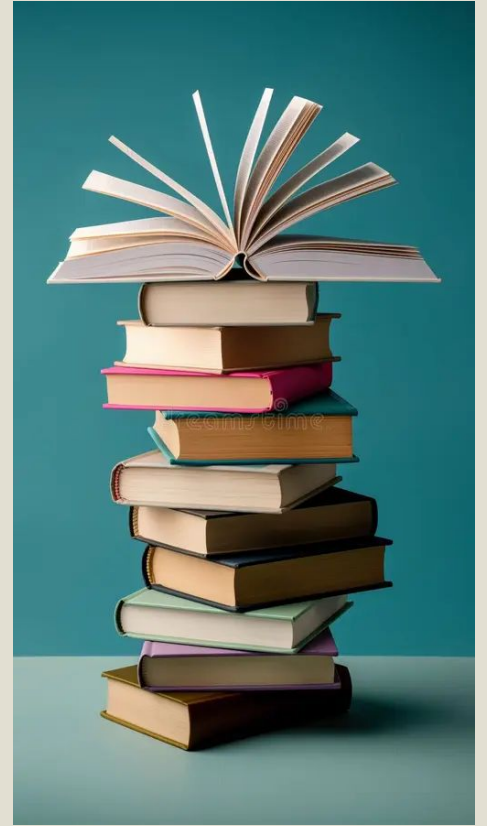


Stack

Undo Mechanism

We implement a "Trash" Stack using LIFO (Last-In-First-Out) logic.

When a meal is deleted, it is pushed onto the stack. Selecting "Undo" pops the item back into the main menu list.

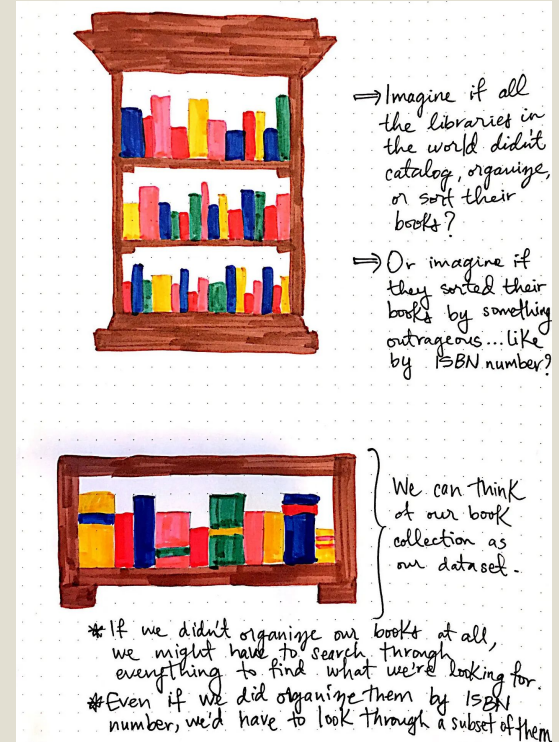


Hash table

Search Optimization

To avoid the $O(n)$ delay of searching through a list, we integrated a Hash Map indexing system.

The meal ID acts as the Key, mapped to the node's memory address. This allows for $O(1)$ average time complexity during searches.



Code Architecture

OOP Principles

Encapsulation: All operations (Add, Edit, Del) are encapsulated within the menuOperation class.

Modularity: Separate classes for Menu logic and Stack management.

Memory Management: Proper destructors ensure no memory leaks occur during runtime.

```
35 class Trach;  
36 class menuOperation{  
37  
38 private:  
39     struct menuLink  
40     {  
41  
42         mealData *meal;  
43         menuLink *link;  
44     } *mLink;  
45  
46     std::unordered_map<int , mealData*> hashmap; //hash table dat  
47  
48 public:  
49     menuOperation(); // for initializ the pointer  
50     void createNewNode_Method(mealData *meal, fstream &Myfile, in  
51     void write_to_file(mealData *meal, fstream &Myfile, int IDMea  
52     void FileToLinkList(mealData *meal, fstream &Myfile);  
53     void addNewMeal(mealData *meal, fstream &Myfile);  
54     void editMealData(mealData *meal, fstream &Myfile, int IDmatc  
55     void SearchMeal(Trach *t, mealData *meal, fstream &Myfile, in  
56     void del(Trach *t, mealData *meal, fstream &Myfile, int IDKey  
57     void displayMenu(mealData *meal);  
58     ~menuOperation();  
59 };  
60  
61 //-----  
62 // Stack  
63 //-----  
64  
65 // Stack for saving the deleted meals  
66 class Trach{
```

Add new meal

```
This file was created successfully!!  
The program is starting ....
```

```
-----Menu operation-----
```

```
[1] Add new meal  
[2] Eddit existing meal information  
[3] Search  
[4] Delete meal for menu  
[5] Undo the delete operation  
[6] Dsiplay the menu  
[7] exit
```

```
Enter the number: 1
```

```
Pleas Enter the meal information:
```

```
Enter ID of the meal: 2
```

```
Enter Name of the meal: pizza
```

```
Enter price of the meal: 7.99
```

```
Enter category of the meal: main
```

```
The new meal was added successfully
```

Display all meals

```
-----Welcome back :)-----
```

```
[1] Add new meal  
[2] Eddit existing meal information  
[3] Search  
[4] Delete meal for menu  
[5] Undo the delete operation  
[6] Dsiplay the menu  
[7] exit
```

```
Enter the number: 6
```

```
Food menu-----
```

```
|ID: 1  Name: burger  price: 4  Category: main
```

```
|ID: 2  Name: pizza  price: 7.99  Category: main
```

```
-----
```

Search for meal

```
-----Welcome back :)-----  
[1] Add new meal  
[2] Eddit existing meal information  
[3] Search  
[4] Delete meal for menu  
[5] Undo the delete operation  
[6] Dsiplay the menu  
[7] exit  
Enter the number: 3  
  
Enter the meal ID: 1  
ID: 1 | Name: burger | price: 4 | Category: main
```

Edit meal ID = 1 info

```
-----Welcome back :)-----  
[1] Add new meal  
[2] Eddit existing meal information  
[3] Search  
[4] Delete meal for menu  
[5] Undo the delete operation  
[6] Dsiplay the menu  
[7] exit  
Enter the number: 2  
  
Enter the meal ID: 1  
  
What is the information you want to edit for a specific meal:  
[1] Edit Name  
[2] Edit Price  
[3] Edit category  
[4] Exit  
Enter the Number: 2  
  
Enter the new price for the meal: 5.50
```

After update the meal ID = 1

```
Food menu-----  
|ID: 1  Name: burger  price: 5.5  Category: main  
|ID: 2  Name: pizza  price: 7.99  Category: main  
-----
```

Delete meal

```
-----Welcome back :)-----  
[1] Add new meal  
[2] Eddit existing meal information  
[3] Search  
[4] Delete meal for menu  
[5] Undo the delete operation  
[6] Dsisplay the menu  
[7] exit  
Enter the number: 4  
  
Enter the meal ID: 1  
ID: 1 | Name: burger | price: 5.5 | Category: main  
  
Do you want to delete this meal [Press Y]: y
```

After the delete operation

```
Food menu-----  
|ID: 2 Name: pizza price: 7.99 Category: main  
-----
```

Undo the deletion

```
-----Welcome back :)-----  
[1] Add new meal  
[2] Eddit existing meal information  
[3] Search  
[4] Delete meal for menu  
[5] Undo the delete operation  
[6] Dsisplay the menu  
[7] exit  
Enter the number: 5  
  
Enter the meal ID: 1
```

After the undo

```
Food menu-----  
|ID: 2 Name: pizza price: 7.99 Category: main  
|ID: 1 Name: burger price: 5.5 Category: main  
-----
```

Summary & Stats:

Total Lines of Code: 650 Lines (Refactored & Optimized).

Data Structures: 3 Integrated Models (Linked List, Stack, Hash Map).

Search Performance: $O(1)$ Average Case.

Program Performance: $O(n)$ Worst Case.

Storage: Persistent CSV-based File System.

THANK YOU